

The Hilbert Uniqueness Method for Internal Control of the One-Dimensional Heat Equation: A Theoretical Companion

Compiled from Cipriano Callejas Hernández's CIMAT MSc thesis
and independently-cited standard results (see text)

Abstract

This document collects, with precise statements and citations, the continuous and discrete controllability theory underlying the Hilbert Uniqueness Method (HUM) applied to *internal* (distributed) control of the one-dimensional heat equation, together with the finite-dimensional (ODE) controllability results that justify passing from the continuous problem to a semi-discrete one. Boundary control is discussed only where it clarifies a genuine point of contrast with the internal case. Results are drawn from Callejas Hernández's 2024 CIMAT thesis, *Numerical and constrained controllability of the heat equation*, whenever available; every result *not* found there is cited independently and marked as such, never attributed to the thesis. Open questions and known limitations are stated explicitly rather than smoothed over.

Contents

1	Well-posedness of the heat equation	2
1.1	Internal (distributed) control	2
1.2	Boundary control	2
2	Regularity of states, controls, and the adjoint state	3
2.1	Internal control	4
2.2	Boundary control	4
2.3	Shared / adjacent	4
3	Controllability definitions	4
4	HUM duality theory	5
4.1	The finite-dimensional (ODE) prototype	5
4.2	Infinite-dimensional generalization	6
5	Penalized HUM: convergence and threshold results	7
5.1	Internal control	8
5.2	Boundary control	8
6	Discrete-in-space uniform controllability: the ODE-controllability bridge	8
6.1	Framing	9
6.2	The space-discrete ODE system	9
6.3	Uniform controllability	10
6.4	Other formulations (brief, not pursued here)	11

7	Known theoretical and numerical issues	11
7.1	Internal control	11
7.2	Boundary control	11
8	Aside: state-positivity and constrained controllability	12

1 Well-posedness of the heat equation

Abstract standing assumption

Let $T > 0$ be the (fixed) control time horizon, and let H and U be real Hilbert spaces – the *state space* and *control space*, respectively (specialized to concrete function spaces below, e.g. $H = U = L^2(0, L)$ for internal control, where $L > 0$ is the length of the spatial interval). Controllability problems are then posed abstractly as

$$y'(t) + Ay(t) = Bu(t) \text{ in } (0, T), \quad y(0) = y_0 \in H, \quad (1)$$

where $u \in L^2(0, T; U)$ is the control, and A, B are (possibly unbounded) linear operators densely defined on $D(A) \subset H$ and $D(B) \subset U$. It is assumed throughout that (1) is *well posed*: for every $(y_0, u) \in H \times L^2(0, T; U)$ there is a unique solution $y \in C([0, T]; H)$ depending continuously on the data, understood in the weak sense via the semigroup generated by $-A$ (Pazy, 2012).

1.1 Internal (distributed) control

For internal control, $H = U = L^2(0, L)$, $A = -\alpha \partial_{xx}$ for a fixed diffusion coefficient $\alpha > 0$, with $D(A) = H^2(0, L) \cap H_0^1(0, L)$, and B is the (bounded) multiplication operator by $\mathbf{1}_\omega$ for a subset $\omega \Subset (0, L)$:

$$\begin{cases} y_t - \alpha y_{xx} = v(x, t) \mathbf{1}_\omega(x), & (x, t) \in (0, L) \times (0, T), \\ y(0, t) = y(L, t) = 0, & t \in (0, T), \\ y(x, 0) = y_0(x), & x \in (0, L). \end{cases} \quad (2)$$

Theorem 1.1 (Well-posedness, internal control). *For every $y_0 \in L^2(0, L)$ and $v \in L^2((0, L) \times (0, T))$, problem (2) has a unique solution $y \in C([0, T]; L^2(0, L))$. Moreover, (2) is null controllable (Andrei Vladimirovich and Oleg Yu, 1996).*

1.2 Boundary control

For boundary control, the same equation is instead driven through a Dirichlet datum at one endpoint:

$$\begin{cases} y_t - \alpha y_{xx} = 0, & (x, t) \in (0, L) \times (0, T), \\ y(0, t) = 0, \quad y(L, t) = v(t), & t \in (0, T), \\ y(x, 0) = y_0(x), & x \in (0, L). \end{cases} \quad (3)$$

Because the control enters through the boundary trace rather than as a bounded source term, (3) only admits a solution *by transposition*, and in a weaker space than the internal case:

Theorem 1.2 (Well-posedness, boundary control). *For every $y_0 \in L^2(0, L)$ and $v \in L^2(0, T)$, problem (3) has a unique weak (transposition) solution $y \in C([0, T]; H^{-1}(0, L))$, and (3) is null controllable (Andrei Vladimirovich and Oleg Yu, 1996).*

From here on, Ω denotes the spatial domain of a result stated at the general (possibly multi-dimensional) level of the underlying PDE-control literature; in the concrete one-dimensional setting of (2)–(3) used throughout the rest of this document, $\Omega = (0, L)$. Correspondingly, $Q := \Omega \times (0, T)$ is the space-time cylinder and $\Sigma := \partial\Omega \times (0, T)$ is its lateral boundary (where the boundary control $v(t)$ of (3) acts).

Remark 1.3. The objectives of exact/null controllability require $y(\cdot, T) \in L^2(\Omega)$, a strictly stronger requirement than the ambient well-posedness class $y \in C([0, T]; H^{-1}(\Omega))$ (Belgacem and Kaber, 2011). This gap between the natural well-posedness space and the space in which controllability is sought is a recurring theme (see the ill-posedness discussion in Section 7).

Solution-by-transposition construction

The rigorous justification for Theorem 1.2 splits the solution as $y = z + w$, where w solves an auxiliary problem with the same data but homogeneous boundary conditions, and z is defined by transposition against a smooth adjoint problem.

Proposition 1.4. *The auxiliary solution w satisfies $w \in L^2(0, T; L^2(\Omega)) \cap C^0([0, T]; H^{-1}(\Omega))$ (Evans, 2010). The adjoint problem*

$$-p_t - \Delta p = f \text{ in } Q, \quad p = 0 \text{ on } \Sigma, \quad p(x, T) = 0 \text{ in } \Omega,$$

with $f \in L^2(0, T; L^2(\Omega))$, has a unique solution $p \in L^2(0, T; H_0^1(\Omega) \cap H^2(\Omega))$ with $p_t \in L^2(Q)$; the transposition identity built from p then yields $z \in C^0([0, T]; H^{-1}(\Omega))$, so $y = z + w$ is well defined (Lions and Magenes, 2012).

This construction is also the continuous-theory root of any numerical HUM variant for boundary control that works directly with the (rougher) transposition solution rather than with the adjoint state.

Remark 1.5 (Gap in the thesis, now filled independently – not from the thesis). No statement of well-posedness in the classical parabolic energy form $y \in C([0, T]; L^2(\Omega)) \cap L^2(0, T; H_0^1(\Omega))$ appears in the thesis; it consistently works either in the abstract semigroup framework above, or in the transposition/weak-solution framework in H^{-1} . For completeness, the standard energy estimate is:

Theorem 1.6 (Classical energy estimate). *For $y_0 \in L^2(\Omega)$ and $f \in L^2(0, T; L^2(\Omega))$, the linear heat equation $y_t - \Delta y = f$ in Q , $y = 0$ on Σ , $y(\cdot, 0) = y_0$, has a unique weak solution $y \in C([0, T]; L^2(\Omega)) \cap L^2(0, T; H_0^1(\Omega))$, with $y_t \in L^2(0, T; H^{-1}(\Omega))$, obtained via the Galerkin method and the energy estimate*

$$\sup_{t \in [0, T]} \|y(t)\|_{L^2(\Omega)}^2 + \int_0^T \|y(t)\|_{H_0^1(\Omega)}^2 dt \leq C \left(\|y_0\|_{L^2(\Omega)}^2 + \|f\|_{L^2(Q)}^2 \right).$$

(Evans, 2010, §7.1). This is the same book already cited in Proposition 1.4 for a different, thesis-attributed result; unlike every thesis-sourced statement in this document (checked line-by-line against the primary source), this statement was not verified page-by-page against Evans' book – it is standard textbook material, not a literal quote.

2 Regularity of states, controls, and the adjoint state

The thesis states no separate, numbered "regularity" theorem for the HUM-optimal control or the state; what follows is what genuinely exists, plus (marked explicitly) what is added independently to complete the picture.

2.1 Internal control

The adjoint final datum lives in $\varphi_T \in L^2(0, 1)$ – no further regularity statement beyond the well-posedness class of Section 1.1 is given.

2.2 Boundary control

The boundary case’s adjoint final datum lives in a strictly smaller space, flagged in the thesis as the key point of departure from the internal case:

Remark 2.1. For boundary control, the adjoint system has final datum $\varphi_T \in H_0^1(0, 1)$, rather than $L^2(0, 1)$ as in the internal case. This single distinction is the source of the H_0^1/H^{-1} machinery needed throughout the boundary theory (Section 1.2, Proposition 1.4).

2.3 Shared / adjacent

The construction of higher-regularity controls (Appendix A.2 of the thesis) is technical machinery specific to the *constrained* (state-positivity) problem of Section 8, not a general HUM control-regularity result for the unconstrained baseline treated here – see that section for the statement (Proposition 8.5). Separately, control regularity for constrained minimal-time controllability is addressed in Lohéac et al. (2017, Thm. 4), again in the constrained setting rather than the baseline.

Remark 2.2 (Gap in the thesis, now filled independently – not from the thesis). No parabolic-smoothing theorem is stated in the thesis, though it is implicit in $D(A) = H^2(\Omega) \cap H_0^1(\Omega)$. Since $A = -\Delta$ generates an analytic semigroup $(e^{-tA})_{t \geq 0}$ on $L^2(\Omega)$ (Pazy, 2012), the uncontrolled solution satisfies the standard regularizing effect

$$y_0 \in L^2(\Omega) \implies y(t) \in D(A) \subset H_0^1(\Omega) \cap H^2(\Omega) \text{ for every } t > 0, \quad \|y(t)\|_{D(A)} \leq \frac{C}{t} \|y_0\|_{L^2(\Omega)}.$$

As with Remark 1.5, this citation is the same book already used for the abstract semigroup framework, and this specific statement was not checked page-by-page against it.

Remark 2.3. The finite-dimensional Kalman-condition/observability machinery underlying *discrete* uniform controllability (Section 6) is a separate notion – “regularity of controllability” for a finite-dimensional ODE system – distinct from the PDE state/control regularity discussed here, and is deliberately not conflated with it.

3 Controllability definitions

These definitions are shared across internal and boundary control – the thesis does not split them, and neither do we. Here H and U are the state and control spaces of (1), as fixed in Section 1.

Definition 3.1 (Controllability notions for (1)). Let $T > 0$. System (1) is said to be:

- (i) **exactly controllable** in time T if, for all $(y_0, y_1) \in H \times H$, there exists $u \in L^2(0, T; U)$ such that the solution satisfies $y(T; y_0, u) = y_1$;
- (ii) **exactly controllable to trajectories** in time T if, for all $(y_0, \hat{y}_0) \in H \times H$ and $\hat{u} \in L^2(0, T; U)$, there exists $u \in L^2(0, T; U)$ such that $y(T; y_0, u) = y(T; \hat{y}_0, \hat{u})$;
- (iii) **null controllable** (or exactly controllable to zero) in time T if, for all $y_0 \in H$, there exists $u \in L^2(0, T; U)$ such that $y(T; y_0, u) = 0$;

- (iv) **approximately controllable** in time T if, for all $(y_0, y_1) \in H \times H$ and every $\varepsilon > 0$, there exists $u \in L^2(0, T; U)$ such that $\|y(T; y_0, u) - y_1\|_H < \varepsilon$.

Equivalently, defining the reachable-set map $\Phi : L^2(0, T; U) \rightarrow H$, $u \mapsto y(T; y_0, u)$, and its range $R(T) = \{y(T; y_0, u) : u \in L^2(0, T; U)\} = \Phi(L^2(0, T; U)) \subset H$, the four notions above correspond to $R(T) = H$, $R(T) \ni y(T; \hat{y}_0, \hat{u})$ for every choice, $0 \in R(T)$, and $R(T)$ dense in H , respectively. (Φ is distinct from the completion space F introduced in Remark 4.5 below – different symbol, different role.)

Remark 3.2. When there is no restriction on the norm of the initial condition, the properties in Definition 3.1 hold *globally*; under such a restriction, controllability holds only *locally*.

Steady-state controllability (constrained-problem aside)

For the constrained/steady-state problem of Section 8 the thesis restates controllability more concretely for the boundary-controlled equation and adds:

Definition 3.3. System (3) is (exactly) controllable to $y_1 \in L^2(\Omega)$ if, for every $y_0 \in L^2(\Omega)$, there exist $T_{y_0} > 0$ and $u \in L^2(\Sigma)$ such that the solution satisfies $y(T_{y_0}) = y_1$.

Definition 3.4 (Steady state). $\bar{y}$ is a steady state of (3) if $-\Delta \bar{y} = 0$ in Ω , $\bar{y} = \bar{u}$ on $\partial\Omega$.

Definition 3.5 (Steady-state controllability). Let y_0, y_1 be steady states. System (3) is exactly-steady-state controllable in time $T > 0$ if there exists $u \in L^2(\Sigma)$ such that the solution satisfies $y(\cdot, 0) = y_0$ and $y(\cdot, T) = y_1$.

4 HUM duality theory

4.1 The finite-dimensional (ODE) prototype

The infinite-dimensional theory below generalizes a finite-dimensional template for the linear ODE system

$$y'(t) = Ay(t) + Bv(t) \text{ in } (0, T), \quad y(0) = y_0, \quad (4)$$

with $n, m \in \mathbb{N}$ (the state and control dimensions), $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $y : [0, T] \rightarrow \mathbb{R}^n$, $v \in L^2(0, T; \mathbb{R}^m)$. This template is stated here in full because it is the direct prerequisite for the discrete/ODE uniform-controllability theory of Section 6: a space semi-discretization reduces the PDE control problem to exactly this ODE setting.

Theorem 4.1 (Kalman rank condition). *Let $K = [B, AB, \dots, A^{n-1}B]$. System (4) is exactly controllable in every time $T > 0$ if and only if $\text{rank } K = n$.*

Source. Coron (2007, Thm. 1.16). □

The adjoint system of (4) is

$$\varphi'(t) = -A^* \varphi(t) \text{ in } (0, T), \quad \varphi(T) = \varphi_T. \quad (5)$$

Theorem 4.2 (Controllability $\Leftrightarrow$ observability). *System (4) is controllable in time T if and only if the adjoint system (5) is observable in time T , i.e. there exists $C = C(T) > 0$ such that, for every solution φ of (5),*

$$|\varphi(0)|_{\mathbb{R}^n}^2 \leq C \int_0^T |B^* \varphi(t)|_{\mathbb{R}^m}^2 dt. \quad (6)$$

Both properties hold in every time T if and only if the Kalman rank condition (Theorem 4.1) holds.

Source. [Zuazua \(2002, Thm. 2.1.1\)](#). □

Theorem 4.3 (HUM control, finite-dimensional). *Let $\mathcal{J} : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathcal{J}(\varphi_T) = \frac{1}{2} \int_0^T |B^* \varphi|^2 + (y_0, \varphi(0))$, where φ solves (5) with $\varphi(T) = \varphi_T$. If $\mathcal{J}$ has a minimizer $\hat{\varphi}_T$, then, letting $\hat{\varphi}$ solve (5) with $\hat{\varphi}(T) = \hat{\varphi}_T$, the control $v = B^* \hat{\varphi}$ is the minimal- L^2 -norm control steering (4) from y_0 to 0 at time T .*

The construction hinges on the *Gramian operator*

$$\Lambda : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad \Lambda(g) := \int_0^T B^* \varphi_g(s) ds, \quad (7)$$

where φ_g solves (5) with final datum g ; Λ is well defined by duality ([Boyer, 2013](#)). This is the same Gramian apparatus used below for the PDE-level construction, over $\mathbb{R}^n$ instead of an infinite-dimensional Hilbert space.

4.2 Infinite-dimensional generalization

In the PDE setting the Kalman rank condition has no direct analogue; more delicate tools are needed, and in general exact controllability of the heat equation fails while approximate controllability holds ([Zuazua, 2002](#)) – see also [Coron \(2007\)](#); [Zabczyk \(2020\)](#) for the general control-theoretic background this document draws on. Approximate controllability of the 1D heat equation can also be established directly via a variational approach ([Fabre et al., 1995](#)), as an alternative to the HUM route developed below. Nonetheless the HUM construction extends: with H, U the state and control spaces of Section 1, and adjoint system in H ,

$$\varphi'(t) = -A^* \varphi(t) \text{ in } (0, T), \quad \varphi(T) = g \in H, \quad (8)$$

Internal control

Proposition 4.4. *The Gramian $\Lambda : H \rightarrow L^2(0, T; U)$ is a positive-definite, symmetric, bounded linear operator ([Boyer, 2013](#)), giving rise to the bilinear form and dual functional*

$$a(g, \xi) = \int_0^T B^* \varphi_g \cdot B^* \varphi_\xi dt, \quad J(g) = \frac{1}{2} a(g, g) + \langle y_0, \varphi_g(0) \rangle, \quad (9)$$

with $\langle \cdot, \cdot \rangle$ the $H \times H'$ duality pairing. For internal control, $H = U = L^2(0, 1)$ and $B^* = B = \mathbf{1}_\omega$.

The thesis develops (9) directly and moves to *penalized* HUM (Section 5) as its route to a well-posed minimization, rather than stating a standalone existence/uniqueness theorem for the unpenalized functional.

Remark 4.5 (Gap in the thesis, now filled independently – not from the thesis). The classical existence/uniqueness argument for the unpenalized infinite-dimensional minimizer is the original Lions completion-space construction. Let $F :=$ the completion of H with respect to the seminorm $\|g\|_F := a(g, g)^{1/2}$. If this seminorm is genuinely a norm on H – equivalent to a unique-continuation property for (8) ($B^* \varphi_g \equiv 0$ on $(0, T) \Rightarrow g = 0$) – then H embeds continuously and densely into F , the linear form $g \mapsto -\langle y_0, \varphi_g(0) \rangle$ extends continuously to F , and Riesz representation on the Hilbert space F yields a unique $\hat{g} \in F$ minimizing J over F . This formalizes what the thesis’s Remark 7.1 (Section 7) describes informally as “the space where J is coercive.” Citation: [Lions \(1988\)](#) – the original HUM source. This is the first citation of Lions’ book *directly* in this document; the thesis itself cites it only indirectly, via [Russell, 1990](#)’s review of it. As with Remarks 1.5 and 2.2, no specific page/theorem number is claimed here.

Theorem 4.6 (Observability inequality, internal control). *If there exists $C > 0$ such that*

$$|\varphi(\cdot, 0)|_{L^2(0,1)}^2 \leq C \int_0^1 \int_0^T |\mathbf{1}_\omega \varphi|^2 dx dt, \quad (10)$$

then J is coercive.

Remark 4.7. The proof of (10) itself requires Carleman estimates and is deferred to [Andrei Vladimirovich and Oleg Yu \(1996\)](#).

Remark 4.8 (Gap in the thesis, now filled independently – not from the thesis: an alternative route). Carleman estimates are not the only route to (10). The Lebeau–Robbiano spectral inequality

$$\|u\|_{L^2(\Omega)}^2 \leq C e^{C\lambda} \|u\|_{L^2(\omega)}^2$$

– valid for any finite combination u of Dirichlet-Laplacian eigenfunctions with eigenvalues $\leq \lambda$, and any open $\omega \subset \Omega$ – yields null controllability of the heat equation from any open subset, in any time $T > 0$, via an iterative/telescoping argument ([Lebeau and Robbiano, 1995](#)). This citation, like [Remarks 1.5, 2.2 and 4.5](#), is standard-form and was not verified page-by-page against the original paper.

Boundary control

Per [Remark 2.1](#), the boundary dual functional $J_b(\varphi_T)$ lives over $\varphi_T \in H_0^1(0, 1)$ rather than $L^2(0, 1)$. No further boundary-specific infinite-dimensional HUM theorem is stated beyond the general development above.

5 Penalized HUM: convergence and threshold results

To obtain a well-posed minimization even when the unpenalized functional J of (9) is not coercive on H , one instead minimizes the penalized functional

$$J_\varepsilon(g) = \frac{1}{2}a(g, g) + \langle y_0, \varphi_g(0) \rangle + \varepsilon \|g\|_H, \quad \varepsilon > 0. \quad (11)$$

Proposition 5.1 (Critical-point inequality). *If J_ε attains its minimum at $\hat{g} \in H$, then for every $\xi \in H$,*

$$a(\hat{g}, \xi) + \langle y_0, \varphi_\xi(0) \rangle \leq \varepsilon \|\xi\|_H,$$

and consequently the controlled solution satisfies $\|y_{\hat{v}}(T)\| \leq \varepsilon$, where $\hat{v} = B^ \hat{\varphi}$.*

Proposition 5.2 (Penalized-HUM duality). *Let $F_\varepsilon(v) := F_1(v) + \frac{1}{2\varepsilon} \|y_v(T)\|_H$ (the primal functional). Then $F_\varepsilon(\hat{v}) = -J_\varepsilon(\hat{g})$, a consequence of the Fenchel–Rockafellar duality theorem ([Ekeland and Temam, 1999](#)) that also holds for $\varepsilon = 0$. Moreover $y(T; y_0, \hat{v}) = -\varepsilon \hat{g}$, $\|y(T; y_0, \hat{v})\|_H \leq \|y(T; y_0, 0)\|_H$, and $\|\hat{g}\|_H \leq \frac{1}{\varepsilon} \|y(T; y_0, 0)\|_H$.*

Source. [Boyer \(2013, Prop. 1.5\)](#). □

Theorem 5.3 (Convergence as $\varepsilon \rightarrow 0$). *Let $y_\varepsilon(T)$ be the state reached by the penalized-HUM control $v_\varepsilon = \hat{v}(\varepsilon)$. Then*

$$y_\varepsilon(T) \rightarrow \mathbb{P}_{Q_F}(y_f(T)) \text{ as } \varepsilon \rightarrow 0,$$

where $Q_F := \{g : B^ \varphi_g \equiv 0 \text{ on } (0, T)\} = \ker \Lambda$ is the set of non-observable states, $\mathbb{P}_{Q_F}$ is the orthogonal projection onto Q_F , and $y_f = y(\cdot; y_0, 0)$ is the free solution. In particular, if $Q_F = \{0\}$ then $y_\varepsilon(T) \rightarrow 0$.*

Source. Boyer (2013, Thm. 1.11). □

Proposition 5.4. (i) *If the primal system is approximately or null controllable, then $y(T; y_0, \hat{v}_\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$.*

(ii) *If the primal system is null controllable, then $v_\varepsilon \rightarrow v^0$ strongly in $L^2(0, T; U)$ as $\varepsilon \rightarrow 0$, where v^0 is the minimal-norm null control (null controllability is needed for the strong convergence, since it guarantees $\inf_{L^2(0, T; U)} F_\varepsilon < \infty$ as $\varepsilon \rightarrow 0$).*

Source. Boyer (2013, Prop. 1.7). □

Remark 5.5. In finite dimension, Q_F is the kernel of the adjoint Kalman matrix K^* of Theorem 4.1; hence $Q_F = \{0\}$ if and only if the Kalman rank condition holds.

5.1 Internal control

Propositions 4.4–5.4 apply directly to internal control over $H = L^2(0, 1)$; nothing further is added beyond this general machinery.

5.2 Boundary control

The same machinery restates over H_0^1/H^{-1} per Remark 2.1. The condition-number result below is stated once, generically, and applies to whichever Gramian is in play (internal or boundary).

Proposition 5.6 (Condition number of the penalized Gramian). *With $\Lambda_\varepsilon := \Lambda + \varepsilon I$, $\|\Lambda_\varepsilon\| = \varepsilon + \|\Lambda\|$ and $\|\Lambda_\varepsilon^{-1}\| = \varepsilon^{-1}$, so the condition number of Λ_ε is*

$$\nu_a = 1 + \frac{1}{\varepsilon} \|\Lambda\|.$$

Theorem 5.7 (Conjugate-gradient convergence rate). *The conjugate-gradient iterates $\{u_n\}_{n \geq 1}$ for minimizing the bilinear form $a(\cdot, \cdot)$ satisfy*

$$\|u_n - u\| \leq C \|u_0 - u\| \left(\frac{\sqrt{\nu_a} - 1}{\sqrt{\nu_a} + 1} \right)^n, \quad n \geq 1,$$

where $\nu_a = \|\Lambda\| \|\Lambda^{-1}\|$ is the condition number of $a(\cdot, \cdot)$: convergence is faster the closer u_0 is to u , and the closer $\sqrt{\nu_a}$ is to 1.

Source. Glowinski (2008). □

Remark 5.8. No further quantitative threshold or error-rate result exists beyond Proposition 5.6 and Theorem 5.7. A scaling law $\varepsilon \sim Ch^p$ (for a mesh-refinement parameter h , diffusion coefficient α , and p larger than the scheme's order of accuracy) is reported, but only as an **experimental conjecture** from numerical observation, not a proven theorem, and is labeled as such wherever referenced.

6 Discrete-in-space uniform controllability: the ODE-controllability bridge

This section covers the theory of *space* semi-discretization – the technique actually used in this work – and its relation to the finite-dimensional (ODE) controllability theory of Section 4.1. Internal (distributed) control is treated as the primary case; boundary control appears only as a one-line contrast.

6.1 Framing

“A space semi-discrete approximation of the parabolic control problems [(3) or (2)] reduces the initial problem to a control problem for an ordinary differential equations (ODE’s) system.”

This reduction is the central bridge exploited throughout this section: study of the semi-discrete heat equation’s controllability becomes an application of Theorems 4.1–4.3 to a specific, mesh-dependent family of linear ODE systems.

The numerical-HUM approach itself originates with the pioneering work of Glowinski et al. (1990) for the wave equation, treated as an unconstrained optimization problem for the adjoint system via convex-duality theory (Ekeland and Temam, 1999); it was later adapted to the heat equation to find null controls by Carthel et al. (1994). HUM belongs to a broader family of dual methods, including the *weighted* HUM, which incorporates a small perturbation to damp oscillatory control behavior (Zuazua, 2006); more recently, primal methods working directly with the primal system via Carleman weights (Fernández-Cara and Münch, 2014) and least-squares/variational methods (Münch and Pedregal, 2014) have given results broadly comparable to the dual approach.

Discrete controllability of parabolic problems is usually studied via three formulations – time-discrete, space-discrete, and fully discrete (Boyer and Hernández-Santamaría, 2020). Uniform controllability (defined precisely in Definition 6.2 below) is established *only* for the space-discrete formulation of the one-dimensional heat equation (Zuazua, 2006; López and Zuazua, 1998); the other two formulations are not pursued in this work (see Section 6.4 for why).

6.2 The space-discrete ODE system

Let $N \in \mathbb{N}$, let $h = L/(N + 1)$ be the corresponding uniform mesh size on $(0, L)$, $x_j = jh$ for $j = 0, \dots, N + 1$, and let $y_h(t) = (y_1(t), \dots, y_N(t)) \in \mathbb{R}^N$ approximate $(y(x_1, t), \dots, y(x_N, t))$ via the standard central-difference approximation of ∂_{xx} . For internal control, (2) becomes the linear ODE system on $\mathbb{R}^N$

$$y'_h(t) = A_h y_h(t) + B_h v_h(t), \quad A_h = \frac{\alpha}{h^2} \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{pmatrix} \in \mathbb{R}^{N \times N}, \quad (12)$$

where $v_h(t) \in \mathbb{R}^N$ and $B_h = \mathbf{1}_{\omega_h} \in \mathbb{R}^{N \times N}$ is the diagonal matrix with $(\mathbf{1}_{\omega_h})_{jj} = 1$ if $x_j \in \omega$ and 0 otherwise (the identity matrix if ω is the whole domain). For boundary control, the analogous system (cf. LeVeque, 2007) has the same $A_h \in \mathbb{R}^{N \times N}$ but $v_h(t) \in \mathbb{R}$ a scalar and $B_h \in \mathbb{R}^{N \times 1}$ a column vector, rather than $v_h(t) \in \mathbb{R}^N$ and $B_h \in \mathbb{R}^{N \times N}$ as in (12).

Proposition 6.1 (Spectrum of A_h). *A_h has eigenvalues $\lambda_j = \frac{2\alpha}{h^2}(1 - \cos(j\pi h))$, $j = 1, \dots, N$, with eigenvectors $e_i^j = \sin(ij\pi h)$, $i = 1, \dots, N$ (LeVeque, 2007, §3.4).*

This explicit, closed-form spectrum is precisely what makes the uniform controllability results below tractable: the proof of Theorem 6.4 “relies solely on the fact that the spectrum of the Laplacian can be computed explicitly.”

6.3 Uniform controllability

Definition 6.2 (Uniform controllability). Controllability properties must be established via an observability inequality for the discrete adjoint system, rather than by directly invoking the Kalman rank condition, since the latter is a **qualitative, binary** criterion for a fixed finite-dimensional system: it says nothing about how the observability constant behaves as the system size grows with $h \rightarrow 0$. System (12) (resp. its boundary analogue) is said to be **uniformly controllable with respect to h** if the discrete observability inequality below holds for a suitable constant $C > 0$ independent of h .

The discrete adjoint system, with $\varphi_h(t) \in \mathbb{R}^N$, is

$$\varphi'_h(t) = -A_h^\top \varphi_h(t) \text{ in } (0, T), \quad \varphi_h(T) = \varphi_h^T \in \mathbb{R}^N. \quad (13)$$

Theorem 6.3 (Discrete observability inequality). *For every $T > 0$ there is $C(T) > 0$ such that*

$$h \sum_{j=1}^N |\varphi_j(0)|^2 \leq C \int_0^T \left| \frac{\varphi_N(t)}{h} \right|^2 dt$$

for every solution φ_h of (13) and every $h > 0$.

Source. López and Zuazua (1998), based on a discrete observability inequality that relies solely on the explicit spectrum of Proposition 6.1. $\square$

Theorem 6.4 (Discrete null controllability and convergence of controls). *For any $T > 0$ and y_h^0 , there is a control $v_h \in L^2(0, T)$ steering the solution of (12) (or its boundary analogue) to $y_j(T) = 0$, $j = 1, \dots, N$. Moreover, for $y^0 \in L^2(0, L)$, the controls v_h may be built so that*

$$v_h \rightarrow v \text{ in } L^2(0, T) \text{ as } h \rightarrow 0, \quad (14)$$

where v is a null control for the continuous heat equation, provided the initial data are chosen appropriately.

Source. López and Zuazua (1998). $\square$

Remark 6.5 (Extension to internal control). Theorem 6.4 – stated by López and Zuazua for the boundary case – can be modified to also give the convergence of the discretization for internal control, system (12) (Zuazua, 2006, Rem. 3.4).

Remark 6.6 (Scope: only in 1D). The uniform property of Definition 6.2 holds *only for the one-dimensional* heat equation (López and Zuazua, 1998); convergence (14) follows from an explicit choice of y^0 via Fourier series (López and Zuazua, 1998, Rem. 1.2); and v_h converges to the minimal-norm HUM control of Section 4.1. In two dimensions, the space semi-discrete heat equation is **not** controllable at all (Zuazua, 2005) – a counterexample that is dimension-general, not specific to internal or boundary control, and that directly explains the 1D restriction here. Further references establishing/refining uniform controllability for the space-discrete 1D case, not worked through in detail here, include Labbé and Trélat (2006) and Boyer et al. (2010).

Remark 6.7 (Gap, not filled: joint (h, k) scaling). Every uniform-controllability result above is for *space* semi-discretization only ($h \rightarrow 0$, time kept continuous); no joint (h, k) -refinement (fully discrete) uniform-controllability theorem is established here. This is the discrete-side counterpart of the open condition-number scaling question already noted after Proposition 5.6: the penalized-HUM condition number $\nu_a = 1 + \|\Lambda\|/\varepsilon$ is stated purely as a function of ε , with no established (h, k) -refinement scaling law. Both gaps should be read together, and neither should be presented as resolved.

6.4 Other formulations (brief, not pursued here)

The time-discrete and fully-discrete formulations are noted only for completeness, since this work uses space discretization exclusively. **Time-discrete:** this is a genuine *negative* result, not a milder version of the space-discrete case above – the time-semi-discrete multi-dimensional heat equation is in general neither null nor approximately controllable (Zheng, 2008), and Carleman estimates for the time-discrete operator give only "another sense of controllability" (Boyer and Hernández-Santamaría, 2020), not the same uniform-observability property. **Fully discrete:** results are scarce; null controllability for a family of one-dimensional parabolic equations has been shown via Carleman inequalities, again only in a relaxed sense of controllability (Casanova and Hernández-Santamaría, 2021).

7 Known theoretical and numerical issues

7.1 Internal control

Remark 7.1 (Ill-posedness of the unpenalized minimization). The space on which J (Proposition 4.4) is coercive is much larger than $L^2(0, 1)$: it is the completion of $L^2(0, 1)$ with respect to the norm $(\frac{1}{2} \int_0^1 \int_0^T |\mathbf{1}_\omega \varphi|^2 dx dt)^{1/2}$ – exactly the space F of Remark 4.5. This space can hardly be approximated by a finite-dimensional basis (Münch and Zuazua, 2010), which is why the numerical problem of minimizing the dual functional is intrinsically ill-posed.

Two genuine cost-of-control results appear in the source material: an exponential-in-*time* estimate, $\|v\|_{L^\infty(\Sigma)} \leq e^{-2\lambda_1 T} C(T - \tau) \|z_0\|_{L^2(\Omega)}$ (here λ_1 is the first Dirichlet-Laplacian eigenvalue on Ω , z_0 the shifted initial datum, and τ an intermediate time – all from the proof of Theorem 8.2 in Section 8, cost in T , not in ε), and a purely *experimental* control-norm blow-up as the control region ω shrinks. No exponential-in- ε blow-up result (of the form $k \sim e^{1/\varepsilon}$, sometimes seen in the wider numerical HUM literature) is established in the source material used here, and none is claimed.

Remark 7.2 (Gramian conditioning vs. control-region geometry). Although the literature guarantees exact/null controllability for internal control on any open ω , with no restriction on its geometry (Zuazua, 2002), the condition number of the Gramian Λ depends strongly on the location of ω ; when discretized in space, Λ_N , it blows up as N increases and ω shrinks (Münch and Periago, 2011).

Remark 7.3 (Time-integrator sensitivity). Convergence of the numerical HUM method is highly sensitive to the choice of time integrator; explicit schemes blow up for most mesh-parameter pairs, even when the underlying finite-difference scheme is convergent and uniformly controllable (Section 6). Gradient-type optimization methods can likewise fail even under those same favorable conditions; preconditioning or other iterative methods for the resulting large linear systems are a natural, and currently active, avenue for improving conjugate-gradient convergence (Theorem 5.7).

Remark 7.4 (Regularity of the adjoint state and the penalization norm). There is a strong dependency between the regularity of the adjoint state and the minimizer of the dual functional: different choices of penalization norm produce genuinely different solution behavior (Kindermann, 1999).

7.2 Boundary control

Boundary control requires a change of norms and function spaces relative to internal control (Remark 2.1), which in practice makes the numerical method more unstable. For a uniform control-point distribution, a denser distribution near the endpoints of the control interval may improve results (Kalimeris et al., 2021).

Remark 7.5 (Lack of robustness). Small perturbations in input data can produce widely different numerical results – a general robustness concern for both control types, not specific to either.

8 Aside: state-positivity and constrained controllability

This section is **related but not implemented** here: it summarizes a separate, constrained-controllability thread from the same source material, included only because two results from it are cited elsewhere in this document (Section 2, Remark 7.1). It is not developed with the same depth as the unconstrained internal-control theory above, and boundary control (the setting in which the thesis develops it) is used here without further internal-control specialization.

Lemma 8.1 (Maximum principle). *The state-positivity constraint reduces to positivity of the boundary control (Pao, 2012, Lem. 4.1).*

Theorem 8.2 (Positivity-constrained steady-state control, linear case). *Let $y_0 \in L^2(\Omega)$, $y_0 \geq 0$, and let $y_1 \in L^2(\Omega)$ be a steady state with $y_0 \neq y_1$ and $y_1 \geq \epsilon$ for some $\epsilon > 0$. Then there exists $T_0 = T_0(y_0, y_1) > 0$ such that, for every $T > T_0$, there is a control $u \in L^2(\Sigma)$ for which the corresponding solution of (3) is non-negative and satisfies $y(\cdot, T) = y_1$.*

The proof shifts $z = y - y_1$ (so $z_0 := z(\cdot, 0) = y_0 - y_1$), uses an observability inequality via Parseval and the first Dirichlet-Laplacian eigenvalue λ_1 (of $-\Delta$ on Ω with homogeneous Dirichlet boundary conditions), invokes the higher-regularity control construction below, and gives the cost estimate $\|v\|_{L^\infty(\Sigma)} \leq C(T)\|z_0\|_{L^2(\Omega)}$, $C(T) = e^{-2\lambda_1 T}C(T-\tau) < 1$ for $T > T_0$ and some intermediate time $\tau \in (0, T)$.

Remark 8.3. A “waiting time” phenomenon governs the minimal achievable time T_0 ; controls achieving exactly T_0 need special regularity (Lohéac et al., 2017, Thm. 4).

Theorem 8.4 (Semilinear staircase method). *Let y_0, y_1 be path-connected bounded steady states with a connecting family $\bar{u}_s \geq \nu > 0$, $s \in [0, 1]$. For T large enough there is $u \in L^\infty(\Sigma)$, $u \geq 0$ a.e., such that the corresponding solution of the semilinear equation satisfies $y(\cdot, T) = y_1$.*

This requires local controllability to trajectories (Pighin and Zuazua, 2018, Lem. 2.1); the staircase method itself originates in Coron and Trélat (2004), building on the first approximate steady-state result of Schmidt (1980). Adjacent results in this thread include semilinear null controllability (Fernández-Cara, 1997; Oleg Yu, 1993), negative results for $n \geq 2$ with time-only controls (Chitour et al., 2005; Martin et al., 2016), and the elliptic regularity ensuring the steady-state set is nonempty (Gilbarg and Trudinger, 1977).

Proposition 8.5 (Hidden parabolic regularity). *Let $N \in \mathbb{N}$, let $\mathcal{O} \subseteq \mathbb{R}^N$ be open (the ambient, possibly extended domain of this proposition – N here is the spatial dimension, unrelated to the mesh-count N of Section 6), let $p \in [1, \infty]$, $\delta \in (0, T)$, and $X^p(\delta, T; \mathcal{O}) := \{u \in L^p(\delta, T; W^{2,p}(\mathcal{O})) : u_t \in L^p((\delta, T) \times \mathcal{O})\}$. For an open $\mathcal{V} \subset \mathcal{O}$ and $\epsilon > 0$, let $\mathcal{V}_\epsilon := \{x \in \mathcal{V} : \text{dist}(x, \mathcal{O} \setminus \bar{\mathcal{V}}) > \epsilon\}$. If $u \in L^2(0, T; H_0^1(\mathcal{O}))$, $u_t \in L^2(0, T; H^{-1}(\mathcal{O}))$ solves $u_t - \Delta u = f$ in $\mathcal{O} \times (0, T)$, $u = 0$ on $\partial\mathcal{O} \times (0, T)$, with $f \in L^2(\mathcal{O} \times (0, T))$, then for $p > 2$: $f \in L^p(\mathcal{V} \times (0, T))$ implies, for every $\epsilon > 0$ and $\delta \in (0, T)$, $u \in X^p(\delta, T; \mathcal{V}_\epsilon)$; and if $p > N + 2$, $u \in L^\infty(\delta, T; W^{1,\infty}(\mathcal{V}_\epsilon))$.*

Source. González-Burgos and de Teresa (2007, Prop. 4.2). □

This proposition – used, via an extended-domain and localization argument, to upgrade an L^2 null control into a higher-regularity control on the control region – is technical machinery specific to this constrained thread, and should not be cited as if it applied to the unconstrained baseline theory of the preceding sections.

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